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Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

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🔗

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This project focuses on aviation accident data to uncover safety trends and risk factors. By analyzing accident severity, aircraft category, phase of flight and level of aircraft damage, it provides insight and recommendation to support safer aviation decisions

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Aviation-Analysis-Project-phase_1



This project focuses on aviation accident data to uncover safety trends and risk factors. By analyzing accident severity, aircraft category, phase of flight and level of aircraft damage, it provides insight and recommendation to support safer aviation decisions.

Overview

The project focuses on analyzing aviation accident data hence it helps a company to decide which aircraft is more cheaper and has a lower risk of fatal accident. The firm is venturing into aviation industry and therefore it requires insights of aviation data in order to make informed decision on which aircraft is best to invest at.

Business Understanding

The stakeholder interested in this project is the Head of the Aviation Division.

Business problem: The company wants to invest in aircraft but they don't know anything about the possible risks of aircraft. Inaccurate aircraft selection might cause very high maintenance, insurance cost and can also lead to more numbers of aircraft death.

Key Business Question: 1. What has changed over time in aircraft safety? 2. Which Aircraft has the lowest number of fatal accidents? 3. Which aircraft makers have the lowest accident rate?

Data Understanding and Analysis

Data Source

-Aviation accident data from 1962-2023 -National Transportation Safety Board

Data preparation

-Filtered for Airplane related Accidents Only -Limited analysis to accidents occurring in the United States - Remove rows with missing aircraft model or make data -Developed indicator for fatal accidents

Visualization of the data

1. Accidents by Aircraft Manufacturer

The bar chart depicts the number of accidents attributed to leading aircraft manufacturers. Some manufacturers tend to have fewer accidents, indicating lower aircraft risk.



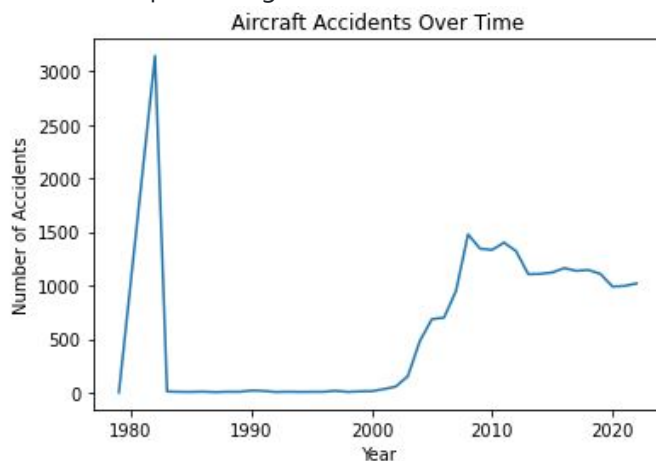
2. Fatal Accident by Aircraft Manufacturer

The visualization shows a comparison of aircraft models based on the percentage of accidents in each model.

Some model of aircraft have a greatly reduced rate of fatal accident rates making them safer investment options for starting.

3. Aircraft Accidents Over Time

A line chart representing the total number of accidents each year. Air safety has improved over time.



It shows Aircraft safety has improved over time, especially in recent years.



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